

group 7

Predicting Early Loan Default risk



Project Goals and Stakeholders

Project overview

Objective: To predict early loan defaults using machine learning. Majorly reducing non-performing loans efficiently, intervene early with high-risk borrowers and Promote financial inclusion through fairer credit decisions

Stakeholders: Key stakeholders include primary stakeholders like credit officers and risk analyst and secondary shareholders like sme lenders and regulators

Data characteristics overview

Insights into loan data features

- The dataset used in this analysis consists of 277,159 loan records collected from Micromart Africa's digital lending platform for the purpose of loan default prediction.
- contains borrower demographics and loan terms.
- The dataset is structured as a binary classification problem, where the target variable indicates whether a borrower defaulted on a loan.

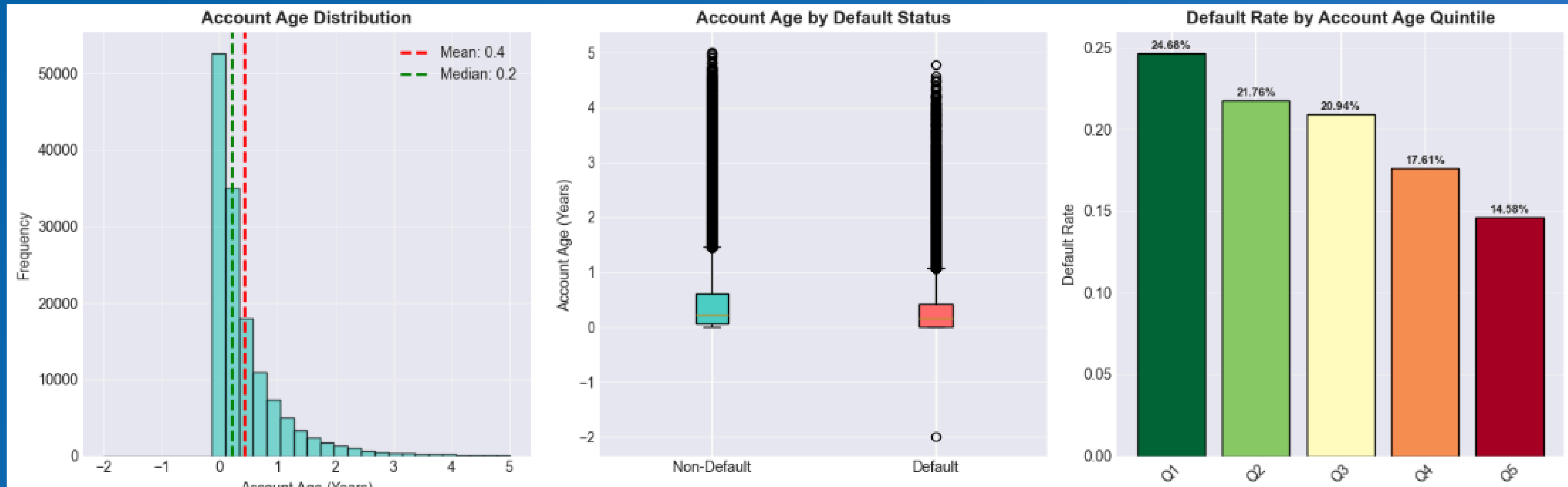
Data Cleaning and Preparation Steps

- The data cleaning process involved handling missing values, standardizing variables and ensuring accurate entries to enhance dataset's overall quality.
- To improve model performance and avoid noise or data leakage, several columns were removed during preprocessing:
 1. Identifiers and system references i.e. loanid
 2. Highly missing features i.e. graduationcount
 3. Operational and administrative fields i.e. approvalstage

This steps ensures the model focuses on meaningful borrower, loan, and repayment behavior features that generalize well for credit risk prediction.

Insights from Exploratory Data Analysis

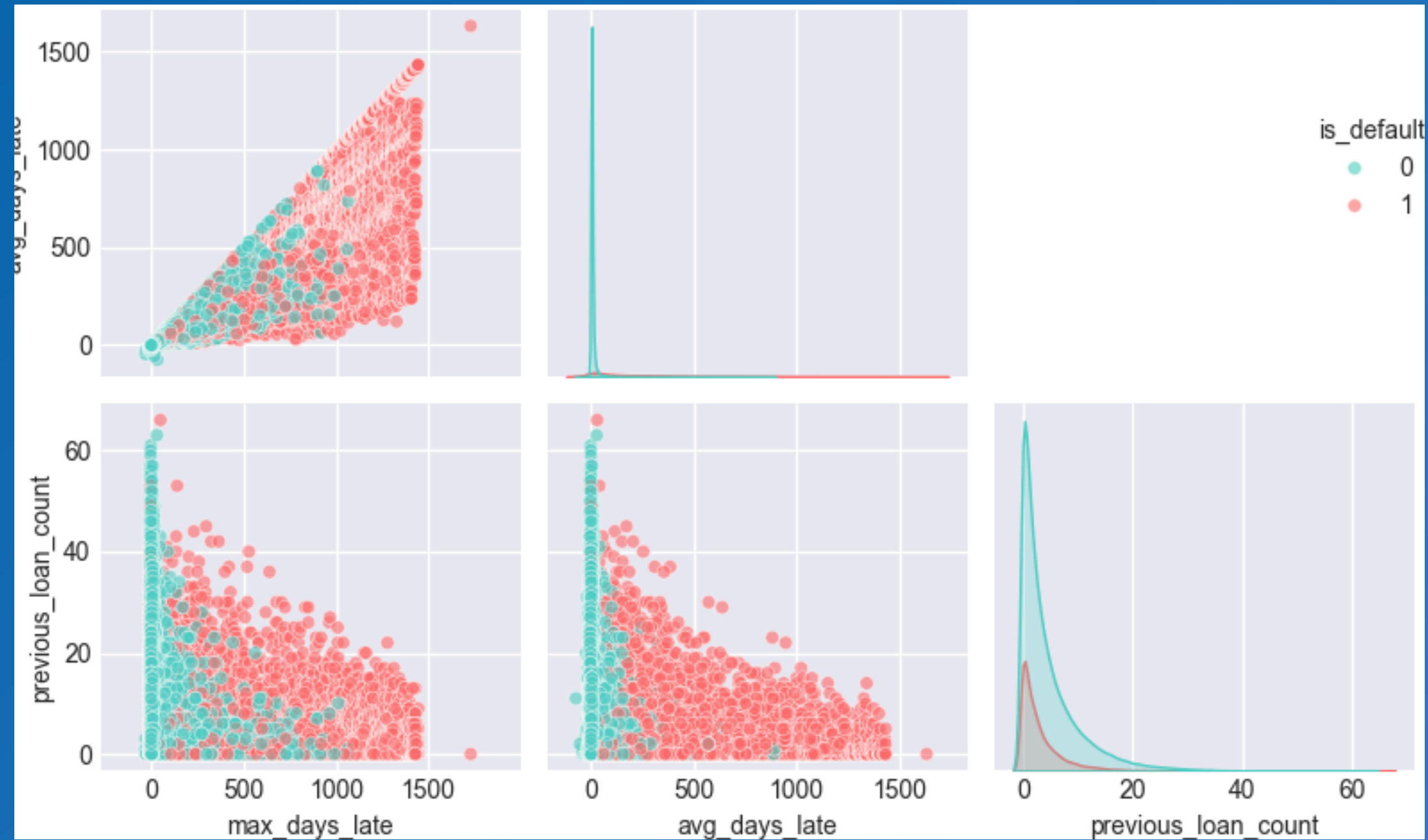
Our EDA revealed that younger entrepreneurs face significantly higher default rates. Additionally, missed payments strongly correlate with eventual defaults.



Age v.s. Default

- We can see that the default risk decreases with age.
- Younger borrowers are significantly riskier.
- correlation with default is -0.07 (weak negative).

Top 3 Variables vs. Default



1. High collinearity between max_days_late and avg_days_late is visually evident.
2. Defaulters have significantly higher days late - clear separation in the scatter plots.
3. previous_loan_count shows weaker discriminatory power visually.
4. Non-linear boundaries may exist - default/non-default separation isn't perfectly linear.

Data Preparation Overview

SCALING

Features were scaled to ensure uniformity, enhancing model performance by allowing algorithms to converge more swiftly and accurately.

ENCODING

Categorical variables were converted into numeric formats, optimizing data processing and enabling machine learning algorithms to interpret the information effectively.

TRAIN-TEST SPLIT

The dataset was divided into training and testing sets, facilitating model evaluation and ensuring its generalizability to unseen data.

Model Performance

Baseline Model Performance

For this we use case:

- Recall is the most critical metric because we want to catch as many potential defaults as possible (minimize false negatives).
- Precision is important but secondary because some false positives are acceptable if we catch more true defaults.

test set results

- Recall: 92.53% → Captures 92.5% of actual defaults (good for early warning)
- Precision: 97.64% → When model predicts "default", it's correct 97.6% of the time
- F1-Score: 95.01% → Strong balance between precision and recall
- Accuracy: 98.07% → Overall prediction accuracy

Advanced Models

Models Chosen

- Random Forest

Why: Captures nonlinear interactions, Robust to noisy features & Strong with behavioral data.

- LightGBM

Why: Employed to model complex non-linear interactions in repayment behavior.

Model Comparison: LightGBM vs. Previous Models

- Our training time showed to evolve with advanced model as logistic regression took 13 minutes while random forest took 2 minutes and LightGBM took one minute

key highlights were:

- Best default detectiona recall because LightGBM achieves the highest recall(95.84%).
- There is balanced performance since get F1-score of 97% and accuracy of 98%.
- The training was efficient due to minimal overfitting and fastest training of 1.5 minutes.

Preferred Model: LightGBM

Why LightGBM is Superior?

1. it has the highest Default Detection Rate (95.84% recall)
2. Best Overall Performance Balance
3. Computational Efficiency
4. Modern Algorithm Advantages

cross-validation results confirm that LightGBM is robust, reliable, and superior for early loan default detection. The high recall (98.70%) directly addresses the business need of catching potential defaults, while the excellent precision (99.64%) ensures operational efficiency.

Recommendations and conclusion

Based on the exploratory analysis and the performance of the predictive models, the following recommendations are proposed to improve credit risk management for SMEs and youth entrepreneurs:

1. Adoption of Machine Learning-Based Credit Scoring in financial institutions instead of only relying on manual credit assessment.
2. Risk-Based Borrower Segmentation to allow the institutions to offer lower interest rates and limit exposure to high risk borrowers.
3. Use of the Model as an Early Warning System to reduce non-performing loans.
4. Integration with Human Decision-Making.
5. Continuous Model Improvement to ensure the model remains relevant time by retraining the model with new data into the model.

THANK YOU FOR
YOUR ATTENTION